



5 – Filters, etc.

XCOPY, FIND, FINDSTR etc.

Search & Find



Simple FIND

◆ Fill **adv4.bat**

```
@echo off
Echo foo > text.txt
Echo Bar >> text.txt
Echo FooBar >> text.txt
FIND /i "Bar" text.txt
Echo -----
FIND /V /n "Foo" text.txt
Echo -----
type text.txt | find /i /n "foo"
Del /Q text.txt
```

Will not display file name

◆ Command **FIND** has some arguments

- **/v** will display lines without match
- **/n** will add line numbers
- Can be used as a filter with « pipe »

FIND inside FOR

◆ Fill adv4.bat

```
:: How many lines with EasterEgg in this script ?
@echo off
set ff=%~f0
set /a eggs=0
for /f "usebackq delims=" %%A in ( `type %%ff%% ^| ^
FIND /i "easteregg" ` ) do (
    set /a eggs+=1
)
echo %eggs% lines containing easterEgg in %~nx0
```



◆ Note multiple escapes

- **usebackq** because **find** string "..."
- **%%ff%%** because in a quoted section
- **^** at end of line to continue command on next line

Command FINDSTR

◆ Fill adv4.bat

```
@echo off
Echo foo > text.txt
Echo Bar >> text.txt
Echo FooBar >> text.txt
type text.txt | findstr ^[fF]
Del /Q text.txt
```

Lines starting with f or F

◆ FINDSTR accepts metacharacters

- . One of Any character
- * Zero or any number of any character
- ^ Start of line
- \$ End of line
- [class] Any character in the group i.e [Aez12]
- [^class] Any character NOT in the group [^ezEZ]
- [x-y] Any character in the interval [0-9]
- \x escape for literal usage of metachar 'x'
- \<xyz « xyz » at start of word
- xyz\> « xyz » at end of word

Command WHERE

◆ Fill adv5.bat

```
@echo off  
  
set /p file_name=Enter pattern for file name:  
  
WHERE /T /R "%LOCALAPPDATA%" "%file_name%"
```

◆ WHERE accepts some metacharacters

- ? One of Any character
- * Zero or any number of any character

Redirections



OUT and IN flows

◆ Fill flow1.bat

```
@echo off
>text.txt echo "FOO"
echo bar >>text.txt
1>>text.txt echo "ACME"
md FOO/BAR 2>log.txt
2>>log.txt md FOO/BAR

echo ----more---- & more text.txt
echo ---- ^< ---- & <text.txt more
echo ----0^< ---- & 0<text.txt more
echo ----FIND---- & <text.txt FIND "ar"
```

◆ Standard flows

- Standard IN is flow 0 (implicit on < or <<)
- Standard OUT is flow 1 (implicit on > or >>)
- ERROR OUT is flow 2

User defined flows

◆ Fill flow2.bat

```
@echo off
1>test.txt echo first line
1>>test.txt echo second line

<&3>LogInput.txt type test.txt
type LogInput.txt
echo.

0<&3>>logInput.txt type test.txt
type LogInput.txt

del /Q LogInput.txt
del /Q test.txt
```

◆ User defined flows

- Use an explicit number i.e **flow 3**
- Use redirection operators **<&3** or **&>3**

Compare



Compare files with FC

◆ Fill adv6.bat

```
@echo off

echo -----BEFORE----- > tempo.txt
Type %~nx0 >> tempo.txt
echo -----AFTER----- >> tempo.txt

FC tempo.txt %~nx0
del /Q tempo.txt
```

◆ FC will give differences between the two files

- Works on files with different sizes
- May receive sets of files
- /L or /U for ASCII or UNICODE text files
- /C to ignore case
- /B for BINARY files
- /LBn to stop after n differences

Compare files with COMP

◆ Fill adv7.bat

```
@echo off

echo -----BEFORE----- > temp1.txt
Type %~nx0 >> temp1.txt
echo -----AFTER----- > temp2.txt
Type %~nx0 >> temp2.txt

COMP temp1.txt temp2.txt /A /M
del /Q temp*.txt
```

◆ COMP will give differences between the two files

- Give details if files have the same
- May receive sets of files
- /A to give the result in ASCII
- /D to give the result in decimal
- /C to ignore ASCII case (A = a)

Copy/Compare with XCOPY

◆ Fill adv8.bat

```
@echo off

Set DIR=%USERPROFILE%\LABS
Set BKP=%USERPROFILE%\BACKUP\LABS
Echo .txt > nsav
Echo .tmp >> nsav

xcopy %DIR%\*.* %BKP%\*.* /D /E /H /I /EXCLUDE:nsav /Y
del /Q nsav
```

◆ XCOPY will update the destination with any new thing

- /D updates files that have been modified
- /E iterates copy through folder tree
- /H copy hidden files
- /I Creates destination folder if necessary
- /EXCLUDE ignore files with a match in nsav (*.txt *.tmp)
- /Y copy without user confirmation